

SankeyLogic Whitepaper & Strategic Overview

Strategic Foundation: A Domain-Independent Framework

SankeyLogic is engineered as a high-performance analytical framework rather than a static reporting tool. By decoupling the **Analytical Layer** from the **Business Logic Layer**, we provide a system that doesn't just draw flows-it models them.

1. The Three Strategic Views

Our engine supports three primary analytical perspectives out of the box, designed to handle extreme structural complexity (up to 15+ levels deep):

A. Network Health & Efficiency (The Diagnostic View)

- **The Goal:** Identify value erosion, structural imbalances, or hidden bottlenecks.
- **Analytical Edge:** The engine automatically detects anomalies in flow integrity and mass balance across the entire network.
- **Application:** In **Finance**, it tracks value leakage across cost centers; in **Energy**, it surfaces conversion losses and system inefficiencies.

B. Impact Simulator (The Predictive View / Downstream)

- **The Goal:** Quantify the "Domino Effect" of local interventions or disruptions.
- **Analytical Edge:** Powered by a sophisticated propagation model, the engine calculates how a change at a single point spreads through connected layers.
- **Application:** **Supply Chain** managers can predict which nodes will feel a disruption 5 steps later; **Sales** leaders can measure the terminal yield of specific lead sources.

C. Upstream Origin (The Attribution View)

- **The Goal:** Trace the DNA of any terminal state back to its system-wide origins.
- **Analytical Edge:** Utilizing real-time graph traversal, the engine calculates **Upstream Footprints**, **Origin Concentration (HHI)**, and **Attribution Metrics** for any selected node.
- **Application:** **Sustainability** officers can trace a product's carbon footprint back to primary energy sources; **Procurement** leads can deconstruct complex cost structures to identify primary price drivers.

2. Modular Architecture & Customization

The system is built to grow with your organizational complexity. Our modular design allows for rapid integration of proprietary logic:

- **Custom Performance Logic:** Implement unique KPIs, specific performance metrics, or proprietary cost-allocation rules as a tailored logic layer.

- **Domain Adaptation:** Whether your data involves financial flows, CO2 tracking, or multi-vector energy systems, the engine adapts to industry standards in a short development cycle.
- **Scalable Intelligence:** New diagnostic panels and specialized analytical "Views" can be integrated without altering the core stability of the system.

3. Enterprise-Grade Architecture

- **Zero-Install (Portable):** Utilizes native technology that requires no administrative privileges, making it instantly deployable in locked-down corporate environments.
- **Data Sovereignty:** 100% offline execution within Excel. Your proprietary data never leaves your local environment-no cloud, no telemetry, no leaks.
- **Deterministic Precision:** Every graph traversal is strictly validated before visualization. What you see is a 100% mathematically accurate reflection of your underlying data model.

Bottom Line

SankeyLogic is designed for environments where "good enough" visualization isn't enough. We provide a customizable analytical standard that turns complex datasets into audit-ready decision intelligence.

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